

第二次作业答案

1. 求图 1 机械臂的末端在基坐标系下的雅可比矩阵

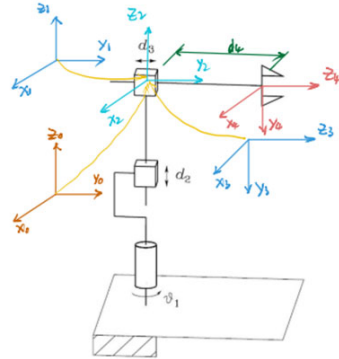


图 1

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	0	d_2	0
3	-90°	0	d_3	0
4	0	0	d_4	0

$${}^0T_1 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1T_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2T_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3T_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_2 = {}^0T_1 {}^1T_2 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_3 = {}^0T_2 {}^2T_3 = \begin{bmatrix} \cos\theta_1 & 0 & -\sin\theta_1 & -\sin\theta_1 d_3 \\ \sin\theta_1 & 0 & \cos\theta_1 & \cos\theta_1 d_3 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_4 = {}^0T_3 {}^3T_4 = \begin{bmatrix} \cos\theta_1 & 0 & -\sin\theta_1 & -\sin\theta_1 d_3 - \sin\theta_1 d_4 \\ \sin\theta_1 & 0 & \cos\theta_1 & \cos\theta_1 d_3 + \cos\theta_1 d_4 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\vec{P} = \begin{bmatrix} -\sin\theta_1 d_4 - \sin\theta_1 d_3 \\ \cos\theta_1 d_4 + \cos\theta_1 d_3 \\ d_2 \end{bmatrix}$$

$$J(\vec{\theta}) = \begin{bmatrix} -\cos\theta_1 d_4 - \cos\theta_1 d_3 & 0 & -\sin\theta_1 \\ -\sin\theta_1 d_4 - \sin\theta_1 d_3 & 0 & \cos\theta_1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

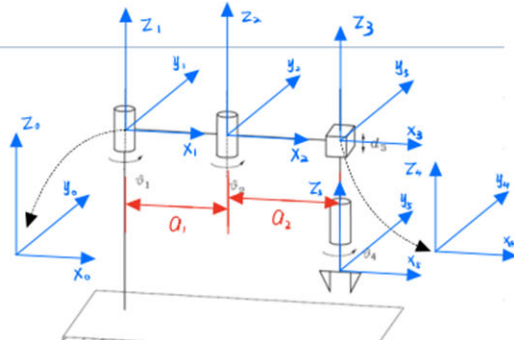
注意：

1. 选取基坐标方向不同, 答案可能不同。如果基坐标 Z 轴朝上的话, 应该只有前两行有所区别。

2. d4 可以为 0

2. 解: a_{i-1} a_i d_i θ_i

	a_{i-1}	a_i	d_i	θ_i
1	0	0	0	θ_1
2	0	a_1	0	θ_2
3	0	a_2	d_3	0
4	0	0	0	θ_4



$${}^0T_1 = \begin{pmatrix} C_1 & -S_1 & 0 & 0 \\ S_1 & C_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1T_2 = \begin{pmatrix} C_2 & -S_2 & 0 & a_1 \\ S_2 & C_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2T_3 = \begin{pmatrix} 1 & 0 & 0 & a_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^3T_4 = \begin{pmatrix} C_4 & -S_4 & 0 & 0 \\ S_4 & C_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^0T_5 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 = \begin{pmatrix} C_1 C_2 - S_1 S_2 & -C_1 S_2 - S_1 C_2 & 0 & C_1 a_1 \\ S_1 C_2 + C_1 S_2 & -S_1 S_2 + C_1 C_2 & 0 & S_1 a_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} C_1 C_2 - S_1 S_2 & -C_1 S_2 - S_1 C_2 & 0 & (C_1 C_2 - S_1 S_2) a_2 + C_1 a_1 \\ S_1 C_2 + C_1 S_2 & -S_1 S_2 + C_1 C_2 & 0 & (S_1 C_2 + C_1 S_2) a_2 + S_1 a_1 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} (C_1 C_2 - S_1 S_2) C_4 - (C_1 S_2 + S_1 C_2) S_4 & -(C_1 C_2 - S_1 S_2) S_4 - (C_1 S_2 + S_1 C_2) C_4 & 0 & (C_1 C_2 - S_1 S_2) a_2 + C_1 a_1 \\ (S_1 C_2 + C_1 S_2) C_4 + (-S_1 S_2 + C_1 C_2) S_4 & -(S_1 C_2 + C_1 S_2) S_4 + (-S_1 S_2 + C_1 C_2) C_4 & 0 & (S_1 C_2 + C_1 S_2) a_2 + S_1 a_1 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} (C_1 C_2 - S_1 S_2) C_4 - (C_1 S_2 + S_1 C_2) S_4 & -(C_1 C_2 - S_1 S_2) S_4 - (C_1 S_2 + S_1 C_2) C_4 & 0 & (C_1 C_2 - S_1 S_2) a_2 + C_1 a_1 \\ (S_1 C_2 + C_1 S_2) C_4 + (-S_1 S_2 + C_1 C_2) S_4 & -(S_1 C_2 + C_1 S_2) S_4 + (-S_1 S_2 + C_1 C_2) C_4 & 0 & (S_1 C_2 + C_1 S_2) a_2 + S_1 a_1 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$J(\vec{\theta}) = \begin{pmatrix} (-S_1 S_2 - C_1 C_2) a_2 - S_1 a_1 & (-C_1 S_2 - S_1 C_2) a_2 & 0 & 0 \\ (C_1 C_2 - S_1 S_2) a_2 + C_1 a_1 & (-S_1 S_2 + C_1 C_2) a_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

$$J(\vec{\theta}) = \begin{pmatrix} -a_1 S_1 - a_2 S_{12} & -a_2 S_{12} & 0 & 0 \\ a_1 C_1 + a_2 C_{12} & a_2 C_{12} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

注意：

3. 选取基坐标方向不同，答案可能不同。如果基坐标 Z 轴朝上的话，应该只有前两行有所区别。
4. 第三行的 1 可以为 -1